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Sang Hyeok Im

Curriculum Vitae

Department of Physics and Astronomy, Seoul National University, 1, Gwanak-ro, Gwanak-gu, Seoul, Republic of Korea — tkdgur0117@gmail.com

EDUCATION

Ph.D student in Astronomy Program (2021.02 ~)

Department of Physics and Astronomy, Seoul National University, 1, Gwanak-ro, Gwanak-gu, Seoul, Republic of Korea

B.S. in Physics (2015.03 ~ 2021.02; double major)

Kyungpook National University, 80, Daehak-ro, Buk-gu, Daegu, 41566, Republic of Korea

B.S. in Astronomy and Atmospheric Sciences (2015.03 ~ 2021.02; double major)

Kyungpook National University, 80, Daehak-ro, Buk-gu, Daegu, 41566, Republic of Korea

PUBLICATIONS

Refereed Journal Article

First Author

1. Testing Lyman Alpha Emitters and Lyman-Break Galaxies as Tracers of Large-Scale Structures at High Redshifts
Sang Hyeok Im, Ho Seong Hwang, et al.,
2024, ApJ, 972, 196

Co-Author

1. ODIN: Identifying Protoclusters and Cosmic Filaments Traced by Ly α -emitting Galaxies
Vandana Ramakrishnan et al., (**Sang Hyeok Im**)
2024, submitted to ApJ (arXiv:2406.08645)
2. The One-hundred-deg² DECam Imaging in Narrowbands (ODIN): Survey Design and Science Goals
Kyoung-Soo Lee et al., (**Sang Hyeok Im**)
2024, ApJ, 962, 36
3. ODIN: Improved Narrowband Ly α Emission Selection Techniques for $z = 2.4, 3.1$, and 4.5
Nicole M. Firestone et al., (**Sang Hyeok Im**)
2023, submitted to ApJ (arXiv:2312.16075)
4. ODIN: Where Do Ly α Blobs Live? Contextualizing Blob Environments within the Large-Scale Structure
Vandana Ramakrishnan, Byeongha Moon, **Sang Hyeok Im** et al.,
2023, ApJ, 951, 119

GRANTS AND FELLOWSHIPS

KIAS Student Fellow (2021.09 ~)

- o Institutional Location: Korea Institute for Advanced Study, 85 Hoegi-ro, Dongdaemun-gu, Seoul 02455, Republic of Korea
- o Project Title: A 100 Deg² DECam Narrow-band Survey for the LSST Era: Tracing the Largest Cosmic Structures in the Distant Universe

PRESENTATIONS

Oral Presentations

1. **9th Galaxy Evolution Workshop**

Date: 2023.02.20 (Mon) ~ 23 (Thu)

Location: Kyoto University, Japan

Title: Testing Lyman alpha emitters and Lyman-break galaxies as tracers of large-scale structures in the high-redshift universe

Poster Presentations

1. **ELT Science in Light of JWST**

Date: 2024 June 3 - 7

Location: Tohoku University, Sendai, Japan

Title: Large-scale structures in the universe traced by Lyman alpha emitters and Lyman-break galaxies

2. **108th KAS Fall Meeting**

Date: 2023 Oct 18 - 20

Location: Ramada Plaza Jeju, Jeju, Republic of Korea

Title: Testing Lyman alpha emitters and Lyman-break galaxies as tracers of the large-scale structures in the high-z universe

3. **XXXIth General Assembly International Astronomical Union**

Date: 2022 Aug 2 - 11

Location: BEXCO, Busan, Republic of Korea

Title: Lyman alpha emitters and Lyman-break galaxies as tracers of large-scale structures in a simulated universe

RESEARCH INTERESTS

1. Large-scale structures of the universe
2. Observational cosmology
3. Environmental effects on galaxy properties
4. Formation and evolution of proto-clusters